CS2810: Mathematics of Data Models

Description

This course teaches the methods and ideas in linear algebra, multivariable calculus, and statistics that are most relevant for the practicing computer scientist doing machine learning, modeling, or hypothesis testing with data. Techniques covered include least squares regression, finding eigenvalues to predict a linear system's behavior, performing gradient descent to fit a model to data, and performing t-tests and chi-square tests to determine whether differences between populations are significant. Includes applications to popular machine learning methods, including Bayesian models and neural networks.

Learning Outcomes

By the end of the course, a student should be able to:
* solve linear systems of equations
* perform least squares regressions
* explain how neural networks "learn" via gradient descent
* solve for eigenvectors and eigenvalues to determine the long-term behavior of linear systems
* find the gradients of basic multivariable functions
* determine whether distributional assumptions such as normality are likely to hold
* test for significance of findings with t-tests and chi-square tests
* know the difference between Bayesian and frequentist statistical approaches

Student Work

Students will submit weekly assignments that are a combination of written work and programming/experimenting in Python, depending on the material. No knowledge of Python would be necessary before the course, and the emphasis is more on using Python as a learning tool for the math than teaching Python per se. The course will have a one midterm and a final exam.

This semester, homework will generally be assigned and due on Fridays.

Instructors

Kevin Gold (Course Coordinator) - kgold@ccs.neu.edu
Matt Higger - mhigger@ccs.neu.edu

Late Policy

You have 5 "late days" to use as you wish over the course of the semester on homework, though no more than 2 can be used on any particular assignment. A late day is an automatic 24-hour extension. You can see how many late days you have left in Blackboard. Our TAs will deduct them automatically on submission. Late days are used in whole number amounts, so a submission the next morning is 1 late day.

Medical absences or absences where you are “representing the school” in some capacity do not count against your late days, as long as you supply us with a note excusing your absence. Other events that are outside your control will be considered on a case-by-case basis.

Course Websites

On Blackboard (blackboard.neu.edu), we will: distribute course notes, distribute and collect assignments, and report grades. Any demo code will also be made available there.

On Piazza (https://piazza.com/northeastern/spring2020/cs2810/home), we will list all staff office hours (Resources->Staff) and offer online Q&A. We have set up the system so that you can choose to post anonymously to other students, but you still aren't anonymous to us (so as to prevent abuse). Abuse of Piazza (i.e. being rude, mean, or simply totally off-topic) could result in loss of Piazza privileges and/or late day privileges.

Textbooks

Since diverse subjects are being covered, online texts are used as a way of keeping student costs down.
Hefferon, *Linear Algebra.* Available free online:
http://joshua.smcvt.edu/linearalgebra/
Lane, *Online Statistics Education: A Multimedia Course of Study*. Available free online:
http://onlinestatbook.com/
Strang, *Calculus.* Available free online:
https://ocw.mit.edu/ans7870/resources/Strang/Edited/Calculus/Calculus.pdf

These won't have material on machine learning or a couple of other application topics; we'll flag those in lecture and in the lecture notes.

Academic Integrity

Students may help each other with homework, but (a) all collaborators (students who gave or received help) must be listed on the homework, and (b) the work shown must be the student’s own, with no dictation or copying from other students (or even from mutual whiteboards). Work or wording that is too similar to be a coincidence will be reported to OSCCR.

Don't lose sight of our focus: preparing you with the skills you'll need to succeed at (and enjoy!) CS work. If you're struggling with coursework, please see us in office hours ASAP. To copy the work of others is to vote against your own abilities. Everyone can succeed but it will take your hard work to get it done!

Grading and Exams

Homework	**45%**
Midterm	**20%**
Final	**35%**

The exams are weighted more because they are our best measure of what you personally have learned from the course. Calculators won't be allowed on the exams, because of the huge range of capabilities of different calculators, but the main equations will be provided.

Homework Collaboration & Internet Use Policies

Discussing problems with your friends is allowed, but copying is not. You may discuss problems and even verbally check your final solutions, but you can't look at another student's assignment (written or code parts). If your assignment does contain sections that are so nearly identical that they could only have been copied, we will file an OSCCR report. (Even math tends to vary from assignment to assignment more than you might think.)

You're encouraged to use programs or resources you find on the Internet as long as you (1) cite the source and (2) in the case of written work, show all work that *would* be necessary to work the problem by hand. In the case of code, you *must* always cite code that is inspired in any way by the Internet, *and* rewrite it "in your own words" as much as is feasible, or risk an OSCCR report. (The exception is using the basic Python documentation, which needs no citation.)

Weekly Topics

Module 1: Linear Models

Week 1
What it means to be linear; equations of planes and hyperplanes; Gauss-Jordan elimination for solving systems of linear equations; working with vectors; intro to Python.

Week 2
Intro to machine learning with perceptrons; rotations and other linear transformations; linear independence; dimension and forming a basis for a space.

Week 3
Subspaces, rank of a matrix, the kernel and losing dimensions in a transformation.

Week 4
Inverses, projections, lines of best fit.

Week 5
Making nonlinear models linear; determinants; similarity; finding eigenvalues and eigenvectors; applications to finding the long-term behavior of linear systems.

Module 2: Multivariable Calculus and Gradient Descent

Week 6
Brief review of calculus; derivatives with respect to variables; directional derivatives; **midterm.**

Week 7
Gradient descent, neural networks, and backpropagation.

Module 3: Probability and Statistics

Week 8
Probability review; notion of probability distribution and variance; introduction to the normal distribution and Poisson distribution.

Week 9
Law of large numbers, central limit theorem, sampling distribution of the mean and standard error; confidence intervals.

Week 10
Biased and unbiased estimates of variance, the t-distribution, hypothesis testing with the t-distribution.

Week 11
Hypothesis testing with the chi-square distribution, covariance and correlation, principal component analysis.

Week 12
Entropy and cross-entropy; introduction to Bayesian graphical models.

Week 13
Bayesian methods continued; review.